AD7031 Optimization Under Uncertainty

Homework 2

Due Oct 31, start of class

Write down your solutions and submit them as a single PDF to `lossof0706@gmail.com`.
Your solutions must be handwritten.

Problem 1. Mean and covariance of a random vector **(25 points)**

Three months of returns are observed, each equally likely ($p_s = \frac{1}{3}$):

	Tech	Energy	Bond
ξ^1	0.00	0.04	0.03
ξ^2	0.06	0.10	0.03
ξ^3	0.12	0.07	0.03

(a) **(10 pts) By hand**, compute the mean $\mu = \mathbb{E}[\tilde{\xi}]$ and the covariance matrix $\Sigma = \mathrm{Cov}(\tilde{\xi})$, and state the dimensions of μ and Σ.

(b) **(10 pts)** Suppose your optimal portfolio is $x^\star = (x_1^\star, x_2^\star, x_3^\star)^\top$. From the table, compute, in terms of $x^\star$,

- the mean $\mathbb{E}[\tilde{\xi}^\top x^\star]$;
- the variance $\mathrm{Var}[\tilde{\xi}^\top x^\star]$.

(c) **(5 pts)** Show that

- $\mathbb{E}[\tilde{\xi}^\top x^\star] = \mu^\top x^\star$;
- $\mathrm{Var}[\tilde{\xi}^\top x^\star] = x^{\star\top} \Sigma\, x^\star$.

hint: we discussed this in class.

Problem 2. Linear program duality **(25 points)**

Consider the following primal linear program and its dual, in the canonical form of the lecture note:

$$\text{(P)} \quad \begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^\top x \\ \text{s.t.} \quad & Ax \le b \end{aligned} \qquad\qquad \text{(D)} \quad \begin{aligned} \max_{\theta \in \mathbb{R}^m} \quad & -b^\top \theta \\ \text{s.t.} \quad & c + A^\top \theta = 0 \\ & \theta \ge 0 \end{aligned}$$

(a) **(8 pts)** Derive the dual problem (D) from the primal problem (P). **Show all the steps clearly.**

(b) **(7 pts)** Consider the primal problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & -4x_1 - 5x_2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \le 60 \\ & x_1 + 3x_2 \le 48 \\ & x_1 \ge 3 \\ & x_1,\ x_2 \ge 0. \end{aligned}$$

- Give c, A, b.
- Write the dual problem (D) numerically.

(c) **(7 pts)** Derive the primal problem (P) from the dual problem (D). **Show all the steps clearly.**

hint: nothing really changes from the derivation in (a).

(d) **(3 pts)** Let $x^\star$ and $\theta^\star$ be optimal solutions of the primal problem (P) and the dual problem (D), respectively. What can you say about $c^\top x^\star$ and $b^\top \theta^\star$?

Problem 3. Conic program duality **(25 points)**

Consider

$$\begin{aligned} \max_{x\in\mathbb{R}^n} \quad & \mu^\top x \\ \text{s.t.} \quad & x^\top \Sigma x \le \sigma^2 \\ & e^\top x \le 1 \\ & x \ge 0 \end{aligned}$$

where $e = [1, 1, \ldots, 1]^\top$ is the all-ones vector and σ is the risk cap.

(a) **(10 pts)** Recall the second-order cone

$$\mathrm{SOC}_Q^{n+1} = \big\{(x,t) \in \mathbb{R}^n \times \mathbb{R} \;:\; \sqrt{x^\top Q\, x} \le t\big\}, \qquad Q \succ 0.$$

Write the model in the conic form $\min c^\top z$ s.t. $Az \le b$, $z \in C$.

- Give z (the decision variable), c, A, b.
- Write C.

hint: use $z = [x,\ t]^\top \in \mathbb{R}^{n+1}$, then write out $c^\top z$ and $Az \le b$ to check that you recover the original problem.

(b) **(10 pts)** With $\Sigma = I$, the dual problem of the second-order cone program in (a) is

$$\begin{aligned} \max \quad & -\theta_1 - \sigma\,\theta_2 \\ \text{s.t.} \quad & \theta_1,\ \theta_2 \ge 0, \quad \theta_3 \in \mathbb{R}^n_+ \\ & \|\theta_1 e - \mu - \theta_3\|_2 \ \le\ \theta_2. \end{aligned}$$

Derive it. **Show all the steps clearly.**

(c) **(5 pts)** State a sufficient condition for strong duality. For which values of σ does it hold for the primal problem in (a)?

Problem 4. Risk measures **(25 points)**

Recall the general stochastic program

$$(\mathrm{SP}) \qquad \begin{aligned} \min_{x\in\mathbb{R}^n} \quad & \mathcal{R}_{\mathbb{P}}\big[f_0(x,\tilde{\xi})\big] \\ \text{s.t.} \quad & \mathcal{R}_{\mathbb{P}}\big[f_i(x,\tilde{\xi})\big] \le 0 \qquad \forall i \in [I], \end{aligned}$$

where $\tilde{\xi} \sim \mathbb{P}$ and $\mathcal{R}_{\mathbb{P}}$ is a risk measure of our choice.

(a) **(5 pts)** Suppose we choose $\mathcal{R}_{\mathbb{P}} = \mathrm{VaR}_\varepsilon$, i.e., the objective function in (SP) is

$$\begin{aligned} \mathrm{VaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big] \ = \ \min_{t\in\mathbb{R}} \quad & t \\ \text{s.t.} \quad & \mathbb{P}\big(f_0(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon. \end{aligned}$$

Using this minimization problem, explain in plain Korean what $\mathrm{VaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big]$ means.

(b) **(5 pts)** Show the reformulation of the constraint $\mathrm{VaR}_\varepsilon\big[f_i(x,\tilde{\xi})\big] \le 0$.

hint: use

$$\mathrm{VaR}_\varepsilon\big[f_i(x,\tilde{\xi})\big] \le 0 \quad\Longleftrightarrow\quad \left[\begin{array}{ll} \min\limits_{t_i\in\mathbb{R}} & t_i \\ \text{s.t.} & \mathbb{P}\big(f_i(x,\tilde{\xi}) \le t_i\big) \ge 1-\varepsilon \end{array}\right] \le 0 \quad\Longleftrightarrow\quad \begin{array}{l} t_i \le 0 \\ \mathbb{P}\big(f_i(x,\tilde{\xi}) \le t_i\big) \ge 1-\varepsilon. \end{array}$$

(c) **(5 pts)** Suppose we choose $\mathcal{R}_{\mathbb{P}} = \mathrm{CVaR}_\varepsilon$, i.e., the objective function in (SP) is

$$\mathrm{CVaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big] \ = \ \min_{\beta\in\mathbb{R}} \ \beta + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{f_0(x,\tilde{\xi}) - \beta,\ 0\}\big].$$

Using this minimization problem, explain in plain Korean what $\mathrm{CVaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big]$ means.

(d) **(5 pts)** Write the reformulation of (SP) with $\mathcal{R}_{\mathbb{P}} = \mathrm{CVaR}_{\varepsilon}$. **Clearly show the reformulation of the objective** $\mathrm{CVaR}_{\varepsilon}\big[f_0(x,\tilde{\xi})\big]$ **and of every constraint** $\mathrm{CVaR}_{\varepsilon}\big[f_i(x,\tilde{\xi})\big] \leq 0$**,** $i \in [I]$**.**

hint: do the same as in (b).

(e) **(5 pts)** For computational tractability, we may prefer CVaR over VaR. Explain why.